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Using Power Query in Microsoft Power BI, Excel, and Fabric

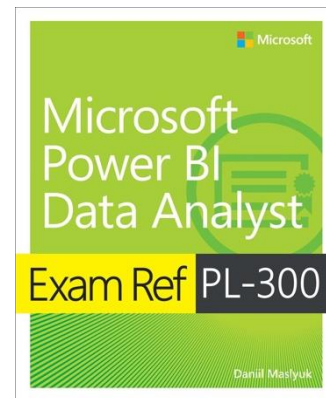
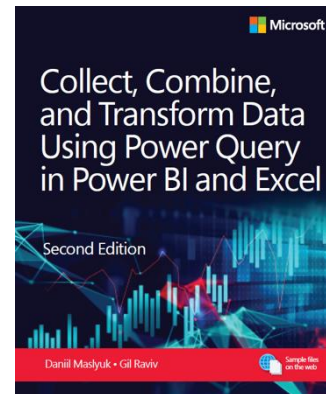
Presented by
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Your presenter today

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Poll: Your Power Query experience

Where do you use Power Query most often?

- Power BI
- Excel
- Fabric



Course overview

- Segment 1: Introducing Power Query
- Segment 2: Transforming Data
- Segment 3: Advanced Concepts



Section 1

Introducing Power Query

- Power Query overview
- Connecting to files, folders, and databases
- Profile the data



Power Query overview

- Power Query is a data extraction and transformation tool
- Power Query is used in Power BI, Excel, and Fabric
- Most actions can be done by using the user interface
- M is the language of Power Query that you can use for advanced transformations



Connecting to files, folders, and databases

- Power Query has over scores of connectors
- You can get data from individual files or whole folders
- When getting data from folders, you'll need to ensure the data format is consistent
- Power Query can fold some operations when getting data from a database

Profile the data

- The user interface provides some profiling information above each column
- Table.Profile can profile whole tables



Section 2

Transforming Data

- Basic transformations
- Appending and merging tables
- Parameters

Basic transformations

- Most transformations can be performed by using the user interface
- Some of the most popular transformations are
 - Filtering
 - Adding columns
 - Removing columns
 - Replacing values

Appending and merging tables

- Appending tables stacks them vertically
- When appending tables, the order of columns doesn't matter
- Merging table combines them horizontally based on key
- You can use more than one column as key to merge tables



Parameters

- You can use parameters in Power Query to change variables in specific places
- Parameters can be reused in different queries
- Parameters can be used when creating Power BI templates



Section 3

Advanced Concepts

- Advanced Editor and M
- Complex types
- Custom functions



Advanced Editor and M

- You can use M to perform advanced operations, such as iterating over a list until finished in a “do while” fashion
- M is a functional case-sensitive language
- You can start writing your own code by changing step formulas in the Formula bar



Complex types

- In Power Query, there are basic types such as text and number, as well as complex types, like lists, records, and tables
- Lists use braces: {1, "a", null}
- Records use brackets: [a = 1, b = 2]
- Complex types can be nested

Custom functions

- You can write custom functions to avoid repeating the same code
- A custom function look like so:

$(X) \Rightarrow X * 2$

- Custom functions can be queries by themselves, written inside other queries, and nested

Thank you!

